THE 'FAITH' FILES



If it's Diwali season, political fireworks surely cannot be far behind. Lighting them is ex-BJP MP **Paresh Rawal,** who plays a travel guide in *The Taj Story*. In that role, he goes to court to question the origins of one of the seven wonders of the world, equating the Mughal-era monument to a "Hindu mahal". The actual courts, meanwhile, asked the filmmakers to tamp down that bit of rhetoric, so the film now comes with a disclaimer. The last political drama that claimed to be "rooted in history"—serial muckraker Vivek Agnihotri's *The Bengal Files*—created barely a ripple at the box office. We'll have to wait and see how this one does.

Lyrical Face-off



After months of the BJP's high-decibel campaign celebrating CM **Pushkar Singh Dhami** as 'Dhakad Dhami', the Opposition Congress has dropped a song of its own. '*Chup re chhoron halla ni kara, dhakad Dhami seenu cha, Uttarakhand ku vikas pracharon ma hunu cha* (Stay quiet boys...Uttarakhand's development is now only in publicity)', takes aim at the 'hollow image' of the state government. The Congress alleges the BJP regime has spent Rs 1,001 crore on self-promotion even as the youth face joblessness and corruption scandals pile up.

PATEL PARADOX

The recent cabinet reshuffle in Gujarat was expected to toss up a Patidar youth icon to counter the rising popularity of AAP MLA Gopal Italia. The names of Hardik Patel and **Jayesh Radadiya** were doing the rounds, but neither made the cut. Hardik is a Congress turncoat, but Radadiya, an outspoken Patel strongman from Saurashtra, has been a BJP loyalist all through. But last year, the 44-year-old had rubbed the party the wrong way by supporting a Patel candidate in a national cooperative body election against the BJP high command's nominee. Worse, his candidate had won. Still, with a Patidar CM at the helm, the community's youth had got their hopes up. It was not to be.



Kaushik Deka with Arkamoy Datta Majumdar, Suhani Singh, Avaneesh Mishra and Jumana Shah

TECH WATCH

By **Ajay Sukumaran**

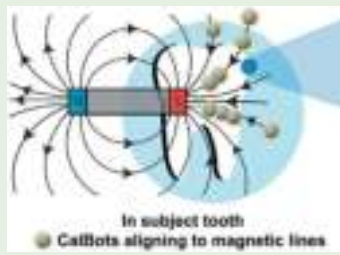
TOOTH SENSITIVITY
NANOBOTS TO THE RESCUE

Switch on the TV and it won't be long before you run into a toothpaste ad for sensitive teeth. Dental hypersensitivity affects nearly one in four people worldwide. It occurs when tooth enamel erodes or the gum recedes, exposing the inner layer of hard tissue, the dentine. Inside the dentine are tiny fluid-filled tunnels, or tubules, which lead to nerve-endings where a twinge is felt when teeth come in contact with cold substances.

Treatments such as desensitising toothpastes provide short-term relief through a protective coating over the exposed part. But a permanent fix remains elusive. Now, a deep-tech startup from the Indian Institute of Science (IISc) in Bengaluru has patented a radically different approach. Using magnets, they guide nano-sized robots deep into the dentinal tubules and plug these tunnels with a bioceramic cement.

The holy grail of fixing sensitive teeth, explains Debayan Dasgupta, director of Theranautilus, which he co-founded with Prof. Ambarish Ghosh of the Centre for Nano Science and Engineering (CeNSE) and postdoc Shanmukh Peddi, has been to drive the cementing material deep enough, given the tubules are about 2,000 microns long. In a paper published in *Advanced Science* in July, the researchers showed that their nanobots—spherical structures with an inner core of iron oxide enclosed by an outer layer of calcium silicate—can reach up to 300-500 microns inside the tubules and, once there, self-assemble into cement-like plugs that block the tubules.





"Forget nanostructures, there are no magnetically-delivered therapeutics anywhere in the world," says Dasgupta. The next step for Theranautilus, proving efficacy in human clinical trials, will start soon.